Lecture 24

Clifford algebras and spin representations

Objectives. *The half-integral fundamental weights of B_n and D_n do not occur in tensor powers of the vector module V. We realise them on an exterior algebra through the Clifford relation.*

By the end of this lecture you will be able to:

- Build the exterior model of the Clifford relation.
- Define the quadratic action of $\mathfrak{so}(V)$ and check its sign.
- Compute weights, parity and dimensions of spin modules.
- Prove irreducibility and identify the highest weights.

Throughout, V is a complex vector space of dimension $N = 2n$ or $N = 2n+1$, with $n \geq 2$. The form q is a nondegenerate symmetric bilinear form on V. As in Lecture 23, $\mathfrak{so}(V)$ consists of the maps $A \in \mathfrak{gl}(V)$ with

$$q(Ax, y) = -q(x, Ay) \quad \text{for all } x, y \in V.$$

24.1 The Clifford relation and the exterior model

Definition 24.1 (Clifford algebra). Let $\mathcal{I} \subseteq T(V)$ be the two-sided ideal generated by all elements $u \otimes v + v \otimes u - 2q(u,v)1$. The *Clifford algebra* is $\mathrm{Cl}(V) = T(V)/\mathcal{I}$. Writing uv for the product of the images of $u, v \in V$,

$$uv + vu = 2q(u,v), \qquad u^2 = q(u,u).$$

A linear map γ from V to a unital algebra that satisfies these relations extends to a unital algebra homomorphism on $T(V)$. This homomorphism vanishes on $\mathcal{I}$, so it factors uniquely through $\mathrm{Cl}(V)$.

Lemma 24.2 (Adapted basis). *The space V has a basis $e_1, \dots, e_n, f_1, \dots, f_n$, together with u_0 when N is odd, such that*

$$q(e_i, e_j) = q(f_i, f_j) = 0, \qquad q(e_i, f_j) = \frac{1}{2}\delta_{ij},$$
$$q(u_0, e_i) = q(u_0, f_i) = 0, \quad q(u_0, u_0) = 1.$$

Proof. Since q is nondegenerate, diagonalising q and rescaling by complex square roots gives an orthonormal basis $x_1, \dots, x_N$. Put

$$e_j = \frac{1}{2}(x_j + i x_{n+j}), \qquad f_j = \frac{1}{2}(x_j - i x_{n+j}), \qquad u_0 = x_{2n+1} \ (\text{odd } N).$$

Then

$$q(e_j,e_j)=q(f_j,f_j)=\frac{1}{4}(1+i^2)=0,\qquad q(e_j,f_j)=\frac{1}{4}(1-i^2)=\frac{1}{2}.$$

Vectors with different indices involve disjoint orthonormal vectors, so q vanishes on them. ∎

Let $W=\operatorname{span}\{e_i\}$ and $W'=\operatorname{span}\{f_i\}$. Both are isotropic of dimension n. An isotropic subspace U satisfies $U\subseteq U^\perp$, so $2\dim U\le N$. Hence W and W' are maximal isotropic.

The model space is $S=\Lambda W$. For $I=\{i_1<\cdots<i_k\}\subseteq\{1,\dots,n\}$ put

$$e_I=e_{i_1}\wedge\cdots\wedge e_{i_k},\qquad e_\emptyset=1.$$

These 2^n vectors form a basis of S.

Definition 24.3 (Creation and annihilation operators). For $w\in W$, the *creation operator* $\mu(w)\colon S\to S$ is given by

$$\mu(w)x=w\wedge x.$$

For $w'\in W'$, the *annihilation operator* $\iota(w')\colon S\to S$ is the linear map with $\iota(w')1=0$ and

$$\iota(w')(w_1\wedge\cdots\wedge w_k)=\sum_{j=1}^{k}(-1)^{j-1}\,2q(w',w_j)\,w_1\wedge\cdots\wedge\widehat{w_j}\wedge\cdots\wedge w_k.$$

Write $\mu_i=\mu(e_i)$ and $\iota_i=\iota(f_i)$.

The right side is multilinear in $w_1,\dots,w_k$. It is also alternating. Suppose $w_a=w_b$ with $a<b$. Terms with $j\neq a,b$ contain a repeated vector. Moving w_a across $b-a-1$ vectors shows that the terms $j=a$ and $j=b$ are one wedge y, with coefficients

$$(-1)^{a-1}c\quad\text{and}\quad(-1)^{b-1}(-1)^{b-a-1}c=(-1)^a c,\qquad c=2q(w',w_a).$$

They cancel, so $\iota(w')$ is well defined on $\Lambda^k W$. Since $2q(f_i,e_j)=\delta_{ij}$,

$$\begin{aligned}\iota_i e_I&=0\quad(i\notin I),\\ \iota_i(e_i\wedge e_J)&=e_J\quad(i\notin J).\end{aligned}$$

Proposition 24.4 (Creation and annihilation relations). *For $w,\tilde w\in W$ and $w',w''\in W'$,*

$$\begin{aligned}\mu(w)\mu(\tilde w)+\mu(\tilde w)\mu(w)&=0,\\ \iota(w')\iota(w'')+\iota(w'')\iota(w')&=0,\\ \mu(w)\iota(w')+\iota(w')\mu(w)&=2q(w',w)\,\mathrm{id}_S\,.\end{aligned}$$

In particular, $\mu_i\iota_j+\iota_j\mu_i=\delta_{ij}$.

Proof. The first relation holds because $w\wedge\tilde w=-\tilde w\wedge w$.

For the third, let $x=w_1\wedge\cdots\wedge w_k$. In $w\wedge x$ the vector w_j sits in position $j+1$. The definition gives

$$\begin{aligned}\iota(w')(w\wedge x)&=2q(w',w)\,x+\sum_{j=1}^{k}(-1)^{j}\,2q(w',w_j)\,w\wedge w_1\wedge\ldots\widehat{w_j}\cdots\wedge w_k\\ &=2q(w',w)\,x-w\wedge\iota(w')x.\end{aligned}$$

This also holds for $x=1$, hence for all $x\in S$ by linearity. It is the third relation.

For the second, let $T = \iota(w')\iota(w'') + \iota(w'')\iota(w')$. Then $T1 = 0$. Put $c' = 2q(w', w)$ and $c'' = 2q(w'', w)$. Two uses of the identity just proved give

$$\begin{aligned}\iota(w')\iota(w'')(w \wedge x) &= \iota(w')\big(c''x - w \wedge \iota(w'')x\big)\\ &= c''\iota(w')x - c'\iota(w'')x + w \wedge \iota(w')\iota(w'')x.\end{aligned}$$

Exchange w' and w'' and add. The terms with coefficients c' and c'' cancel, so

$$T(w \wedge x) = w \wedge Tx.$$

Induction on degree gives $T = 0$. ∎

Theorem 24.5 (Exterior model). *Let $\theta\colon S \to S$ be the* parity operator*, given by $\theta x = (-1)^k x$ for $x \in \Lambda^k W$. For $w \in W$, $w' \in W'$ and $t \in \mathbb{C}$ define*

$$\begin{aligned}\gamma(w + w') &= \mu(w) + \iota(w') \qquad (N = 2n),\\ \gamma(w + w' + tu_0) &= \mu(w) + \iota(w') + t\theta \quad (N = 2n+1).\end{aligned}$$

Then $\gamma(u)\gamma(v) + \gamma(v)\gamma(u) = 2q(u,v)\,\mathrm{id}_S$ for all $u, v \in V$. Hence γ extends to an algebra homomorphism $\gamma\colon \mathrm{Cl}(V) \to \mathrm{End}(S)$.

Proof. Both sides are symmetric and bilinear in (u, v). By polarisation it suffices to prove $\gamma(u)^2 = q(u,u)$. For $u = w + w'$, Proposition 24.4 gives

$$\gamma(u)^2 = \mu(w)^2 + \iota(w')^2 + \mu(w)\iota(w') + \iota(w')\mu(w) = 2q(w', w).$$

Isotropy of W and W' gives

$$q(u,u) = q(w,w) + 2q(w,w') + q(w',w') = 2q(w,w').$$

In the odd case, θ anticommutes with $\mu(w)$ and $\iota(w')$, because these change degree by one. Also $\theta^2 = 1$. So the cross terms vanish, and

$$\gamma(w + w' + tu_0)^2 = \gamma(w + w')^2 + t^2\theta^2 = 2q(w,w') + t^2 = q(w + w' + tu_0,\, w + w' + tu_0).$$ ∎

24.2 Quadratic elements and the action of $\mathfrak{so}(V)$

For $u, v \in V$ define $M_{u,v} \in \mathfrak{gl}(V)$ by

$$M_{u,v}x = 2q(v,x)u - 2q(u,x)v.$$

Lemma 24.6 (Bivectors and skew maps). *Each $M_{u,v}$ lies in $\mathfrak{so}(V)$. The map $u \wedge v \mapsto M_{u,v}$ is a linear isomorphism $\Lambda^2 V \to \mathfrak{so}(V)$. For $A \in \mathfrak{so}(V)$,*

$$[A, M_{u,v}] = M_{Au,v} + M_{u,Av}.$$

Proof. The expression

$$q(M_{u,v}x, y) = 2q(v,x)q(u,y) - 2q(u,x)q(v,y)$$

is antisymmetric in (x, y), so $M_{u,v} \in \mathfrak{so}(V)$.

Take an orthonormal basis $x_1, \dots, x_N$. Then

$$M_{x_k,x_l}x_m = 2\delta_{lm}x_k - 2\delta_{km}x_l,$$

so the matrix of M_{x_k,x_l} is $2(E_{kl}-E_{lk})$. In this basis $\mathfrak{so}(V)$ is the space of skew matrices, with basis $E_{kl}-E_{lk}$ for $k<l$. The map sends the basis $x_k\wedge x_l$ $(k<l)$ of $\Lambda^2 V$ to a basis of $\mathfrak{so}(V)$.

For the bracket, use $q(v,Ax)=-q(Av,x)$ and $q(u,Ax)=-q(Au,x)$:

$$\begin{aligned}[A,M_{u,v}]x &= 2q(v,x)Au-2q(u,x)Av-2q(v,Ax)u+2q(u,Ax)v\\ &= \big(2q(v,x)Au-2q(Au,x)v\big)+\big(2q(Av,x)u-2q(u,x)Av\big)\\ &= M_{Au,v}x+M_{u,Av}x. \qquad \blacksquare\end{aligned}$$

We fix this identification of bivectors with skew maps. Define the linear map $\sigma\colon \mathfrak{so}(V)\to \mathrm{Cl}(V)$ by

$$\sigma(M_{u,v})=\frac{1}{2}(uv-vu)=uv-q(u,v).$$

It is well defined because the right side is bilinear and alternating in (u,v).

Theorem 24.7 (The quadratic action)**.** *For $A,B\in\mathfrak{so}(V)$ and $x\in V$, in $\mathrm{Cl}(V)$:*

$$\text{(a)}\quad [\sigma(A),x]=Ax, \qquad\qquad \text{(b)}\quad [\sigma(A),\sigma(B)]=\sigma([A,B]).$$

Hence $\rho=\gamma\circ\sigma\colon \mathfrak{so}(V)\to\mathfrak{gl}(S)$ is a representation, and $[\rho(A),\gamma(x)]=\gamma(Ax)$.

Proof. (a) Apply the Clifford relation twice:

$$\begin{aligned}uvx &= u\big(-xv+2q(v,x)\big)\\ &= -\big(-xu+2q(u,x)\big)v+2q(v,x)u\\ &= xuv-2q(u,x)v+2q(v,x)u.\end{aligned}$$

Thus

$$[uv,x]=M_{u,v}x.$$

The scalar $q(u,v)$ is central, so $[\sigma(M_{u,v}),x]=M_{u,v}x$. Linearity gives (a) for all A.

(b) Let $B=M_{u,v}$. The rule $[a,bc]=[a,b]c+b[a,c]$ and (a) give

$$\begin{aligned}[\sigma(A),\sigma(B)] &= [\sigma(A),uv]\\ &= [\sigma(A),u]v+u[\sigma(A),v]\\ &= (Au)v+u(Av)\\ &= \sigma(M_{Au,v})+\sigma(M_{u,Av})+q(Au,v)+q(u,Av)\\ &= \sigma([A,B]).\end{aligned}$$

The last step uses the skewness $q(Au,v)=-q(u,Av)$ and Lemma 24.6. Linearity gives (b) for all B. The algebra homomorphism γ preserves commutators, which gives the final assertions. $\blacksquare$

We now compute Cartan actions and a bracket. The signs depend on the factor 2 in $M_{u,v}$ and the order uv in σ. Put $H_i=M_{e_i,f_i}$. Since $2q(f_i,e_j)=\delta_{ij}$,

$$H_ie_j=\delta_{ij}e_j, \qquad H_if_j=-\delta_{ij}f_j, \qquad H_iu_0=0.$$

The definition of σ gives

$$\sigma(H_i)=e_if_i-q(e_i,f_i)=e_if_i-\frac{1}{2}.$$

On S, write $N_i=\mu_i\iota_i$. Then

$$\rho(H_i)=\gamma(e_i)\gamma(f_i)-\frac{1}{2}=N_i-\frac{1}{2}.$$

For a bracket, take $A = M_{e_1,e_2}$ and $B = M_{f_1,f_2}$. The definition of $M_{u,v}$ gives

$$Af_1 = -e_2, \qquad Af_2 = e_1.$$

Lemma 24.6 and $M_{f_1,e_1} = -H_1$ give

$$[A, B] = M_{Af_1,f_2} + M_{f_1,Af_2} = M_{-e_2,f_2} + M_{f_1,e_1} = -H_1 - H_2.$$

Since $q(e_1, e_2) = q(f_1, f_2) = 0$,

$$\rho(A) = \mu_1\mu_2, \qquad \rho(B) = \iota_1\iota_2.$$

By Proposition 24.4 with $\delta_{12} = 0$, operators with index 1 anticommute with those with index 2. The same proposition gives $\iota_i\mu_i = 1 - N_i$. Hence

$$\begin{aligned}[\mu_1\mu_2, \iota_1\iota_2] &= \mu_1\mu_2\iota_1\iota_2 - \iota_1\iota_2\mu_1\mu_2 \\ &= -\mu_1\iota_1\mu_2\iota_2 + \iota_1\mu_1\iota_2\mu_2 \\ &= -N_1N_2 + (1 - N_1)(1 - N_2) = 1 - N_1 - N_2 = \rho(-H_1 - H_2).\end{aligned}$$

24.3 Weights, parity and dimensions

Let $\mathfrak{h} = \operatorname{span}\{H_1, \dots, H_n\}$, and let $\varepsilon_i \in \mathfrak{h}^*$ satisfy $\varepsilon_i(H_j) = \delta_{ij}$. By the formulas for H_i above, e_i, f_i and u_0 have weights ε_i, $-\varepsilon_i$ and 0. For weight vectors x, y of weights a, b, Lemma 24.6 gives

$$[H, M_{x,y}] = M_{Hx,y} + M_{x,Hy} = (a + b)(H)\, M_{x,y}.$$

The $M_{x,y}$ for unordered pairs of distinct basis vectors form a basis of $\mathfrak{so}(V)$. So $\mathfrak{h}$ is the zero-weight space, and the root vectors are

vector	M_{e_i,e_j}	M_{f_i,f_j}	M_{e_i,f_j} $(i \neq j)$	M_{e_i,u_0}	M_{f_i,u_0}
root	$\varepsilon_i + \varepsilon_j$	$-\varepsilon_i - \varepsilon_j$	$\varepsilon_i - \varepsilon_j$	ε_i	$-\varepsilon_i$
on S	$\mu_i\mu_j$	$\iota_i\iota_j$	$\mu_i\iota_j$	$\mu_i\theta$	$\iota_i\theta$

with $i < j$ in the first two columns, and the last two columns only for N odd. Each pair has $q(x, y) = 0$, so $\rho(M_{x,y}) = \gamma(x)\gamma(y)$. This gives the last row. The positive roots of Lecture 23 are $\varepsilon_i \pm \varepsilon_j$ $(i < j)$, and also ε_i for B_n. Use the fundamental weights of Lecture 23. For D_2, set

$$\omega_1 = \frac{1}{2}(\varepsilon_1 - \varepsilon_2), \qquad \omega_2 = \frac{1}{2}(\varepsilon_1 + \varepsilon_2).$$

Proposition 24.8 (Weights and parity).

(a) The vector e_I has weight

$$\nu_I = \frac{1}{2}\Big(\sum_{i \in I} \varepsilon_i - \sum_{i \notin I} \varepsilon_i\Big).$$

The weights of S are the 2^n weights $\frac{1}{2}(\pm\varepsilon_1 \pm \cdots \pm \varepsilon_n)$, each of multiplicity one.

(b) If $N = 2n$, the spaces

$$S^+ = \bigoplus_{k \equiv n \ (2)} \Lambda^k W, \qquad S^- = \bigoplus_{k \not\equiv n \ (2)} \Lambda^k W$$

are submodules of dimension 2^{n-1}. The weights of S^+ have an even number of minus signs. The weights of S^- have an odd number.

Proof. (a) If $i \in I$, write $e_I = \pm e_i \wedge e_{I\setminus\{i\}}$. Then $\iota_i e_I = \pm e_{I\setminus\{i\}}$ and μ_i restores e_I, so $N_i e_I = e_I$. If $i \notin I$, then $\iota_i e_I = 0$. Hence

$$\rho(H_i)e_I = \frac{1}{2}e_I \qquad (i \in I),$$
$$\rho(H_i)e_I = -\frac{1}{2}e_I \quad (i \notin I).$$

Distinct subsets give distinct sign patterns.

(b) For basis vectors $x, y \in W \cup W'$, the operator $\rho(M_{x,y}) = \gamma(x)\gamma(y) - q(x,y)$ changes degree by -2, 0 or 2. These operators span $\rho(\mathfrak{so}(V))$, so parity is preserved. The number of minus signs in ν_I is $n - |I|$, which is even exactly on S^+. The map $I \mapsto I\triangle\{1\}$ exchanges even and odd subsets, so each half has 2^{n-1} elements. ∎

The weights of S have coordinates in $\frac{1}{2} + \mathbb{Z}$. Weights of tensor powers of V lie in $\mathbb{Z}^n$ (Lecture 23). Hence no tensor power of V has a submodule isomorphic to S (odd N) or $S^\pm$ (even N).

24.4 Irreducibility and highest weights

Theorem 24.9 (Spin and half-spin modules)**.**

(a) If $N = 2n$, then $S^+ \cong L(\omega_n)$ and $S^- \cong L(\omega_{n-1})$. Both are irreducible of dimension 2^{n-1}.

(b) If $N = 2n+1$, then $S \cong L(\omega_n)$, irreducible of dimension 2^n.

Proof. Step 1: a weight vector. Let $U \neq 0$ be a submodule of $S^\pm$ (even case) or of S (odd case). The commuting diagonalisable operators $\rho(H_i)$ preserve U, so U is the sum of its intersections with weight spaces. The weight spaces of S are the lines $\mathbb{C}e_I$. Hence $e_I \in U$ for some I.

Step 2: moving between basis vectors. The operators in the last row of the root table send each basis vector to ± a basis vector or to 0:

$$\mu_i\mu_j e_I = \pm e_{I\cup\{i,j\}} \qquad (i \neq j;\ i,j \notin I),$$
$$\iota_i\iota_j e_I = \pm e_{I\setminus\{i,j\}} \qquad (i \neq j;\ i,j \in I),$$
$$\mu_i\iota_j e_I = \pm e_{(I\setminus\{j\})\cup\{i\}} \quad (j \in I,\ i \notin I).$$

In the even case, let e_J be a basis vector with $|J| \equiv |I|$ modulo 2. Removing pairs of indices takes e_I to $\pm e_\emptyset$ or to some $\pm e_{\{k\}}$. In the second case, $|J|$ is odd, so $J \neq \emptyset$. If $k \notin J$, apply $\mu_l\iota_k$ with $l \in J$. Adding pairs then gives $\pm e_J$. Hence $U = S^\pm$.

In the odd case, $\iota_i\theta$ removes an index $i \in I$ and $\mu_i\theta$ adds an index $i \notin I$, up to sign. These reach $e_\emptyset$ and then every e_J. Hence $U = S$.

Step 3: highest vectors. The positive root vectors act by

$$\mu_i\mu_j, \quad \mu_i\iota_j \ \ (i<j), \qquad \mu_i\theta \ \ (B_n \text{ only}).$$

Each operator kills $e_{[n]} = e_1 \wedge \cdots \wedge e_n$. After any contraction, the creation step repeats an index.

On $e_{[n-1]} = e_1 \wedge \cdots \wedge e_{n-1}$, the operator $\mu_i\mu_j$ needs two absent indices, but only n is absent. For $\mu_i\iota_j$, either $j = n$ and $\iota_n e_{[n-1]} = 0$, or $j < n$ and i is still present. So $\mathfrak{n}_+ e_{[n-1]} = 0$ in the even case.

Step 4: identification. If an irreducible module L has a highest vector v of weight λ, the universal property of $M(\lambda)$ gives a nonzero map $M(\lambda) \to L$. It is onto because L is simple.

The unique-simple-quotient theorem of Lecture 12 gives $L \cong L(\lambda)$. Apply this to $e_{[n]}$ of weight ω_n, which lies in S^+ or in S, and to $e_{[n-1]} \in S^-$ of weight ω_{n-1}. Proposition 24.8 gives the dimensions. ∎

24.5 Worked example: the half-spin modules of $\mathfrak{so}_6$

Take $n = 3$. Then

$$S^+ = \Lambda^3 W \oplus \Lambda^1 W, \qquad S^- = \Lambda^0 W \oplus \Lambda^2 W.$$

Order the basis of S^+ as (e_{123}, e_1, e_2, e_3). Its weights are

$$\begin{aligned} e_{123}&: \ \frac{1}{2}(1,1,1),\\ e_1&: \ \frac{1}{2}(1,-1,-1),\\ e_2&: \ \frac{1}{2}(-1,1,-1),\\ e_3&: \ \frac{1}{2}(-1,-1,1). \end{aligned}$$

The basis $(1, e_{23}, e_{13}, e_{12})$ of S^- has the negatives of these weights, in the same order.

The simple roots are

$$\alpha_1 = \varepsilon_1 - \varepsilon_2, \qquad \alpha_2 = \varepsilon_2 - \varepsilon_3, \qquad \alpha_3 = \varepsilon_2 + \varepsilon_3.$$

Since $(\alpha_1, \alpha_2) = (\alpha_1, \alpha_3) = -1$ and $(\alpha_2, \alpha_3) = 0$, the diagram is the A_3 chain $\alpha_3 - \alpha_1 - \alpha_2$. On $(\alpha_3, \alpha_1, \alpha_2)$, the labels of ω_3 and ω_2 are $(1,0,0)$ and $(0,0,1)$. These are the two ends of the chain.

The simple root vectors for $\alpha_3, \alpha_1, \alpha_2$ are M_{e_2,e_3}, M_{e_1,f_2} and M_{e_2,f_3}. On S^+ they act by $\mu_2\mu_3$, $\mu_1\iota_2$ and $\mu_2\iota_3$. Their only nonzero values on the basis are

$$\begin{aligned} \mu_2\mu_3 e_1 &= e_2 \wedge e_3 \wedge e_1 = e_{123},\\ \mu_1\iota_2 e_2 &= \mu_1 1 = e_1,\\ \mu_2\iota_3 e_3 &= \mu_2 1 = e_2. \end{aligned}$$

In the ordered basis (e_{123}, e_1, e_2, e_3), the matrices of the raising operators are matrix units of $\mathfrak{sl}_4$. Write $D(A)$ for the matrix of $\rho(A)$ in this basis:

$$D(M_{e_2,e_3}) = E_{12}, \qquad D(M_{e_1,f_2}) = E_{23}, \qquad D(M_{e_2,f_3}) = E_{34}.$$

Proposition 24.10 ($\mathfrak{so}_6$ and $\mathfrak{sl}_4$). *The restriction $\rho_+ \colon \mathfrak{so}(V) \to \mathfrak{gl}(S^+)$ of ρ to S^+ is a Lie algebra isomorphism. Through it, S^+ is the defining $\mathfrak{sl}_4$-module and $S^- \cong (S^+)^*$.*

Proof. The algebra $\mathfrak{so}_6$ is simple, since its diagram is connected (Lecture 10). So it equals its derived algebra (Lecture 4), and ρ_+ lands in the traceless operators. Its kernel is a proper ideal, hence zero. Both sides have dimension 15.

The dual $(S^+)^*$ is irreducible. Its weights are the negatives $-\nu$ of the weights ν of S^+. By the weight lists above, these are exactly the weights of S^-. The highest weight of $L(\lambda)$ is its unique maximal weight. Equal weight sets therefore give $S^- \cong (S^+)^*$ by Lecture 13. ∎

Self-checks

(1) What are the dimensions and highest weights of the spin modules of $\mathfrak{so}_8$ and $\mathfrak{so}_9$?
(2) Why is $e_\emptyset = 1$ not a highest vector of S?
(3) Which half-spin module of $\mathfrak{so}_{10}$ contains e_{13}, and what is its weight?

Answers. (1) $\mathfrak{so}_8$: $S^+ \cong L(\omega_4)$ and $S^- \cong L(\omega_3)$, both of dimension 8. $\mathfrak{so}_9$: $S \cong L(\omega_4)$, of dimension 16. (2) Because $\rho(M_{e_1,e_2})1 = \mu_1\mu_2 1 = e_{12} \neq 0$. (3) Here $n = 5$ and $2 \not\equiv 5$ modulo 2, so $e_{13} \in S^-$. Its weight is $\frac{1}{2}(1,-1,1,-1,-1)$.

Summary

Creation and annihilation operators on ΛW realise the Clifford relation. The elements $\frac{1}{2}(uv - vu)$ give the action of $\mathfrak{so}(V)$, with half-integral weights $\frac{1}{2}(\pm\varepsilon_1 \pm \cdots \pm \varepsilon_n)$.

For D_n, $S = S^+ \oplus S^- \cong L(\omega_n) \oplus L(\omega_{n-1})$. For B_n, $S \cong L(\omega_n)$.

Reading

W. Fulton and J. Harris, *Representation Theory: A First Course*, §20.1. R. Carter, *Lie Algebras of Finite and Affine Type*, §13.5. Goodman–Wallach, *Symmetry, Representations, and Invariants*, Chapter 6.